In [1]:
```
!pip install gensim
```

```
Requirement already satisfied: gensim in C:\Users\manis\AppData\Local\Python\pyth
oncore-3.12-64\Lib\site-packages (4.4.0)
Requirement already satisfied: numpy>=1.18.5 in C:\Users\manis\AppData\Local\Pyth
on\pythoncore-3.12-64\Lib\site-packages (from gensim) (2.5.4)
Requirement already satisfied: scipy>=1.7.0 in C:\Users\manis\AppData\Local\Pytho
n\pythoncore-3.12-64\Lib\site-packages (from gensim) (1.18.1)
Requirement already satisfied: smart_open>=1.8.1 in C:\Users\manis\AppData\Local
\Python\pythoncore-3.12-64\Lib\site-packages (from gensim) (9.0.1)
Requirement already satisfied: wrapt in C:\Users\manis\AppData\Local\Python\pytho
ncore-3.12-64\Lib\site-packages (from smart_open>=1.8.1->gensim) (2.5.2)
```

In [2]:
```
# a. Data preparation

# Import necessary packages
import re
from gensim.models import Word2Vec

# Read the text document
file = open('document3.txt', 'r', encoding='utf-8')
text = file.read()
file.close()

print("Original Text:")
print(text)

# Convert text to lowercase
text = text.lower()

# Remove special characters and numbers
text = re.sub(r'[^a-z\s]', '', text)

# Tokenize the text into words
words = text.split()

print("\nTokenized Words:")
print(words)

# Prepare sentences for Word2Vec
sentences = [words[i:i+5] for i in range(len(words)-4)]

# b. Train Word2Vec CBOW Model
model = Word2Vec(
    sentences=sentences,
    vector_size=50,
    window=3,
    min_count=1,
    sg=0,
    epochs=100
)

print("\nCBOW Model Training Completed!")

# c. Evaluate Results

# Display vocabulary
print("\nVocabulary:")
print(model.wv.index_to_key)
```

```
# Display word embedding
word = 'influenza'

print("\nWord Embedding of", word)
print(model.wv[word])

# Find similar words
print("\nSimilar Words:")
print(model.wv.most_similar(word, topn=5))

# Check similarity between two words
print("\nSimilarity between COVID and Influenza:")
print(model.wv.similarity('covid', 'influenza'))

# Save the trained model
model.save('document3_cbow.model')

print("\nModel saved successfully!")
```

```
Original Text:
The speed of transmission is an important point of difference between the two vir
uses. Influenza has a shorter median incubation period (the time from infection t
o appearance of symptoms) and a shorter serial interval (the time between success
ive cases) than COVID-19 virus. The serial interval for COVID-19 virus is estimat
ed to be 5-6 days, while for influenza virus, the serial interval is 3 days. This
means that influenza can spread faster than COVID-19.

Further, transmission in the first 3-5 days of illness, or potentially pre-sympto
matic transmission –transmission of the virus before the appearance of symptoms –
is a major driver of transmission for influenza. In contrast, while we are learni
ng that there are people who can shed COVID-19 virus 24-48 hours prior to symptom
onset, at present, this does not appear to be a major driver of transmission.

The reproductive number – the number of secondary infections generated from one i
nfected individual – is understood to be between 2 and 2.5 for COVID-19 virus, hi
gher than for influenza. However, estimates for both COVID-19 and influenza virus
es are very context and time-specific, making direct comparisons more difficult.

Tokenized Words:
['the', 'speed', 'of', 'transmission', 'is', 'an', 'important', 'point', 'of', 'd
ifference', 'between', 'the', 'two', 'viruses', 'influenza', 'has', 'a', 'shorte
r', 'median', 'incubation', 'period', 'the', 'time', 'from', 'infection', 'to',
'appearance', 'of', 'symptoms', 'and', 'a', 'shorter', 'serial', 'interval', 'th
e', 'time', 'between', 'successive', 'cases', 'than', 'covid', 'virus', 'the', 's
erial', 'interval', 'for', 'covid', 'virus', 'is', 'estimated', 'to', 'be', 'day
s', 'while', 'for', 'influenza', 'virus', 'the', 'serial', 'interval', 'is', 'day
s', 'this', 'means', 'that', 'influenza', 'can', 'spread', 'faster', 'than', 'cov
id', 'further', 'transmission', 'in', 'the', 'first', 'days', 'of', 'illness', 'o
r', 'potentially', 'presymptomatic', 'transmission', 'transmission', 'of', 'the',
'virus', 'before', 'the', 'appearance', 'of', 'symptoms', 'is', 'a', 'major', 'dr
iver', 'of', 'transmission', 'for', 'influenza', 'in', 'contrast', 'while', 'we',
'are', 'learning', 'that', 'there', 'are', 'people', 'who', 'can', 'shed', 'covi
d', 'virus', 'hours', 'prior', 'to', 'symptom', 'onset', 'at', 'present', 'this',
'does', 'not', 'appear', 'to', 'be', 'a', 'major', 'driver', 'of', 'transmissio
n', 'the', 'reproductive', 'number', 'the', 'number', 'of', 'secondary', 'infecti
ons', 'generated', 'from', 'one', 'infected', 'individual', 'is', 'understood',
'to', 'be', 'between', 'and', 'for', 'covid', 'virus', 'higher', 'than', 'for',
'influenza', 'however', 'estimates', 'for', 'both', 'covid', 'and', 'influenza',
'viruses', 'are', 'very', 'context', 'and', 'timespecific', 'making', 'direct',
'comparisons', 'more', 'difficult']

CBOW Model Training Completed!

Vocabulary:
['the', 'of', 'for', 'virus', 'covid', 'influenza', 'transmission', 'to', 'is',
'and', 'a', 'are', 'days', 'be', 'than', 'interval', 'serial', 'between', 'numbe
r', 'driver', 'major', 'in', 'can', 'that', 'this', 'while', 'symptoms', 'appeara
nce', 'from', 'time', 'shorter', 'viruses', 'making', 'timespecific', 'context',
'very', 'both', 'estimates', 'however', 'higher', 'understood', 'individual', 'in
fected', 'one', 'generated', 'infections', 'secondary', 'reproductive', 'appear',
'not', 'does', 'present', 'at', 'onset', 'symptom', 'prior', 'hours', 'shed', 'wh
o', 'people', 'there', 'learning', 'we', 'contrast', 'before', 'presymptomatic',
'potentially', 'or', 'illness', 'first', 'further', 'faster', 'spread', 'means',
'estimated', 'cases', 'successive', 'infection', 'period', 'incubation', 'media
n', 'has', 'two', 'difference', 'point', 'important', 'an', 'direct', 'comparison
s', 'more', 'speed', 'difficult']

Word Embedding of influenza
[-2.21680030e-02 -7.52550159e-06 -1.47534400e-01  3.10139470e-02
```

```
 -1.10441312e-01 -1.34644806e-01  2.93856740e-01  1.28438503e-01
 -1.99724749e-01 -1.93533644e-01  4.27911505e-02 -3.68947476e-01
  4.17550743e-01  2.06256300e-01 -2.79504180e-01  1.21063791e-01
  1.55179605e-01  2.30335563e-01 -3.35725367e-01 -1.73690856e-01
  1.82236463e-01  2.10692838e-01  4.47968602e-01  6.00919016e-02
  2.21320450e-01  1.56840757e-02 -4.72633950e-02  2.78522104e-01
 -2.42304236e-01  2.39956845e-02 -8.85012373e-02 -1.89109460e-01
 -4.73609678e-02 -2.07560942e-01 -1.01801850e-01  3.24464068e-02
  1.91737235e-01 -8.33416581e-02 -3.98968719e-02  5.69000505e-02
  1.28426403e-01  1.07557159e-02  1.00519694e-01 -7.17823282e-02
  4.54180121e-01 -7.58052543e-02 -8.11682865e-02 -1.93789721e-01
  8.88582990e-02 -4.85499427e-02]

Similar Words:
[('while', 0.9928766489028931), ('than', 0.9928374290466309), ('viruses', 0.99230
91530799866), ('and', 0.991992712020874), ('however', 0.9914198517799377)]

Similarity between COVID and Influenza:
0.9903465

Model saved successfully!
```

In [ ]: